

Research Interests

3D Computer Vision, Image and Video Signal Processing, Machine Learning

Education

Ph.D. - Indian Institute of Science (IISc), Bengaluru Aug 2018 - May 2024

Dept of Electrical Communication Engineering

- ◊ Advisor : Dr. Rajiv Soundararajan
- ◊ CGPA : 9.59 (PMRF Fellow)

B.E. - PES Institute of Technology, Bengaluru Aug 2012 - May 2016

Dept of Electronics and Communication Engineering

- ◊ CGPA : 9.93 (Gold Medallist)

Class XII : 96% (District Topper) 2012

Class X : 96.8% (District Topper) 2010

Course Work

Mathematics : Linear Algebra, Probability, Optimization

Electrical : Digital Image Processing, Computer Vision, Machine Learning for Signal Processing, Digital Video Processing, Detection and Estimation Theory

Professional Experience

- ◊ **SpreeAI Corporation**, Research Scientist Aug 2026 - Present
Working on Sparse View 4D reconstruction of dynamic scenes using 3DGS and video diffusion models.
- ◊ **SpreeAI Corporation**, Research Scientist May 2024 - June 2026
Created the first known prototype on size-accurate deformable 3D mesh of a garment from a single flat-garment image.
- ◊ **PES University**, Visiting Faculty Jan 2021 - May 2023
- ◊ **Indian Institute of Science (IISc)**, Teaching Assistant Oct 2020 - May 2022
- ◊ **Cisco Systems India Pvt Ltd**, Software Engineer Aug 2016 - Jul 2018
- ◊ **Elseem Inc.**, Research Intern Jun 2015 - Jul 2015

Selected Publications

- ◊ Nagabhushan Somraj, Sai Harsha Mupparaju, Adithyan Karanayil, and Rajiv Soundararajan. Simple-RF: Regularizing sparse input radiance fields with simpler solutions. *ACM Transactions on Graphics (TOG)*, 45(3):1–24, 2026.
- ◊ Gurutva Patle, Nilay Girgaonkar, Nagabhushan Somraj, and Rajiv Soundararajan. AD-GS: Alternating densification for sparse-input 3d gaussian splatting. In *ACM SIGGRAPH Asia*, December 2025.
- ◊ Kapil Choudhary, Nagabhushan Somraj, and Rajiv Soundararajan. Bidirectional flow fields for sparse input novel view synthesis of dynamic scenes. In *IEEE International Conference on Image Processing (ICIP)*, 2025.
- ◊ Nagabhushan Somraj, Kapil Choudhary, Sai Harsha Mupparaju, and Rajiv Soundararajan. Factorized motion fields for fast sparse input dynamic view synthesis. In *ACM SIGGRAPH*, 2024.
- ◊ Nagabhushan Somraj and Rajiv Soundararajan. ViP-NeRF: Visibility prior for sparse input neural radiance fields. In *ACM SIGGRAPH*, 2023.

Academic Service

- ◇ Reviewer: CVPR, ICCV, ECCV, AACL, WACV, BMVC, ISMAR, TIP, TCSVT, TCI.
- ◇ **Outstanding Reviewer:** ECCV 2024, WACV 2024, ISMAR 2022.

Honors & Awards

- ◇ Finalist in IKDD Doctoral Dissertation in Data Science Award 2025.
- ◇ **Best Presentation Award** at EECS Symposium 2024, IISc.
- ◇ Recipient of Prime Minister's Research Fellowship (PMRF): 2020, 2021, 2022.
- ◇ Recipient of MHRD scholarship (Govt of India): 2012, 2013, 2014, 2015.